Human Reproduction

2.0 Reproductive Events in Humans

As you are aware, **humans are sexually reproducing and viviparous**. Human reproduction runs through a fixed series of events, each with its own name.

1. **Gametogenesis -** the formation of gametes: **sperms** in males and **ovum** in females.
2. **Insemination -** the **transfer of sperms into the female genital tract**.
3. **Fertilisation -** the **fusion of male and female gametes**, leading to formation of the **zygote**.
4. Formation and development of the **blastocyst** and its **attachment to the uterine wall (implantation)**.
5. **Gestation -** the embryonic development inside the uterus.
6. **Parturition -** the delivery of the baby.

All these reproductive events **occur only after puberty**. There is an important difference between the sexes in how long gamete formation lasts: **sperm formation continues even in old men**, **but formation of ovum ceases in women around the age of fifty years**.

2.1 The Male Reproductive System

The **male reproductive system is located in the pelvis region**. It includes a **pair of testes** along with **accessory ducts**, **glands** and the **external genitalia**.

2.1a The Testes

The **testes are situated outside the abdominal cavity** within a pouch called the **scrotum**. The scrotum **maintains the low temperature of the testes - 2-2.5°C lower than the normal internal body temperature -** which is **necessary for spermatogenesis**.

In adults, **each testis is oval**, with a **length of about 4 to 5 cm** and a **width of about 2 to 3 cm**. The testis is **covered by a dense covering**. Each testis has **about 250 compartments called testicular lobules**, and **each lobule contains one to three highly coiled seminiferous tubules** in which sperms are produced.

Each **seminiferous tubule** is lined on its inside by **two types of cells**: the **male germ cells (spermatogonia)** and the **Sertoli cells**. The **male germ cells undergo meiotic divisions**, finally leading to **sperm formation**, while the **Sertoli cells provide nutrition to the germ cells**.

The regions **outside the seminiferous tubules**, called **interstitial spaces**, contain **small blood vessels** and **interstitial cells**, also called **Leydig cells**. The **Leydig cells synthesise and secrete testicular hormones called androgens**. Other **immunologically competent cells are also present**.

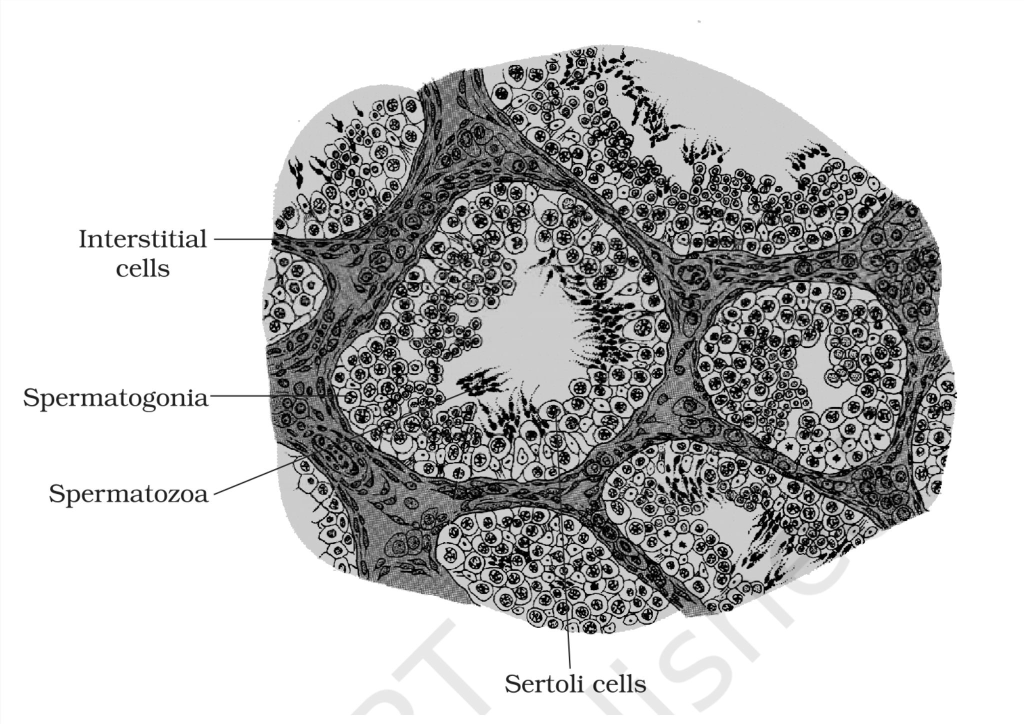

Fig. 2.2 - Diagrammatic sectional view of a seminiferous tubule. Labelled parts: interstitial cells, spermatogonia, spermatozoa and Sertoli cells.

2.1b Accessory Ducts, Penis and Glands

The **male sex accessory ducts** include the **rete testis**, **vasa efferentia**, **epididymis** and **vas deferens**. Their connected path carries the sperms outward:

1. The **seminiferous tubules** of the testis open into the **vasa efferentia** through the **rete testis**.
2. The **vasa efferentia leave the testis** and open into the **epididymis**, located **along the posterior surface of each testis**.
3. The **epididymis leads to the vas deferens**, which **ascends to the abdomen and loops over the urinary bladder**.
4. The vas deferens **receives a duct from the seminal vesicle** and opens into the **urethra** as the **ejaculatory duct**.
5. The **urethra originates from the urinary bladder** and **extends through the penis** to its external opening called the **urethral meatus**.

Together, **these ducts store and transport the sperms** from the testis to the outside through the urethra.

The **penis is the male external genitalia**. It is made up of a **special tissue that helps in erection of the penis to facilitate insemination**. The **enlarged end of the penis, called the glans penis, is covered by a loose fold of skin called the foreskin**.

The **male accessory glands** include the **paired seminal vesicles**, a **prostate** and the **paired bulbourethral glands**. The **secretions of these glands constitute the seminal plasma**, which is **rich in fructose, calcium and certain enzymes**. The **secretions of the bulbourethral glands also help in the lubrication of the penis**.

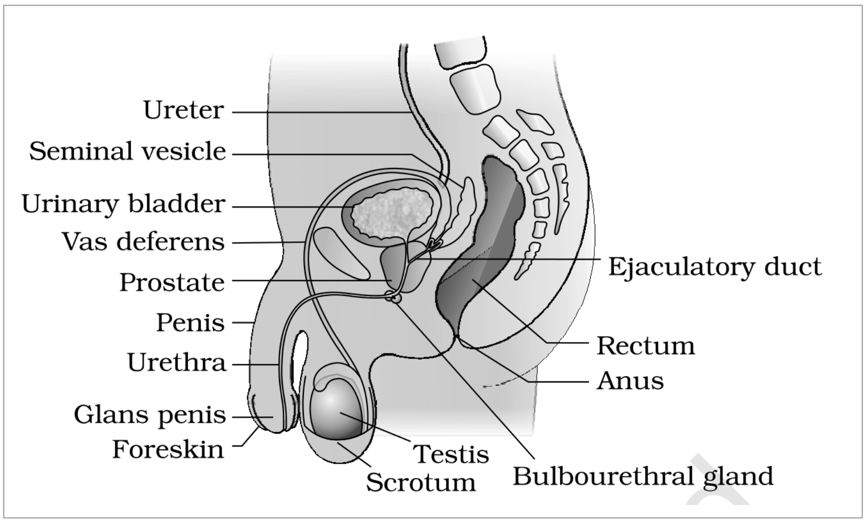

Fig. 2.1a - Diagrammatic sectional view of the male pelvis showing the reproductive system. Labelled parts: ureter, urinary bladder, seminal vesicle, ejaculatory duct, prostate, bulbourethral gland, vas deferens, urethra, penis, glans penis, foreskin, testis, scrotum, rectum and anus.

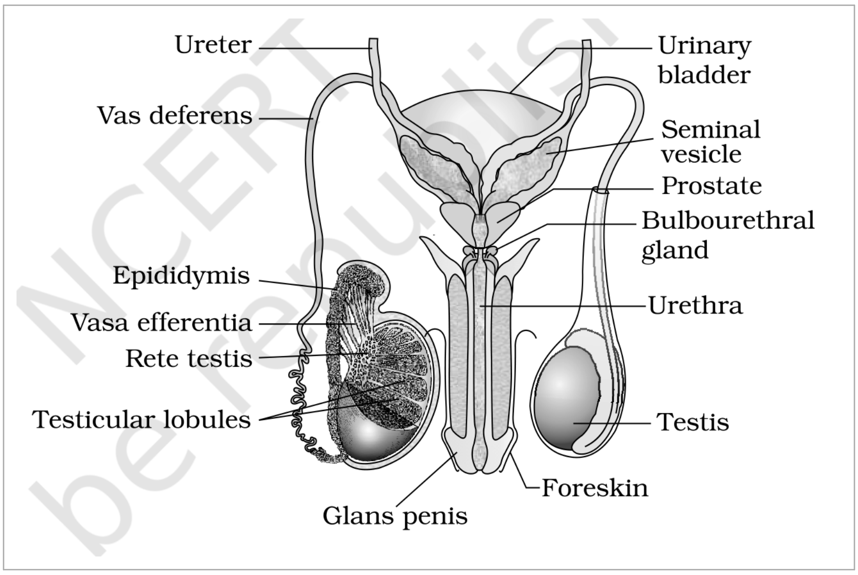

Fig. 2.1b - Diagrammatic view of the male reproductive system (part of the testis is opened to show inner details). Labelled parts: ureter, urinary bladder, seminal vesicle, prostate, bulbourethral gland, vas deferens, epididymis, vasa efferentia, rete testis, testicular lobules, urethra, glans penis, foreskin and testis.

2.2 The Female Reproductive System

The **female reproductive system** consists of a **pair of ovaries** along with a **pair of oviducts**, a **uterus**, a **cervix**, a **vagina** and the **external genitalia**, located in the **pelvic region**. These parts, along with a **pair of mammary glands**, are **integrated structurally and functionally to support ovulation, fertilisation, pregnancy, birth and child care**.

2.2a The Ovaries

The **ovaries are the primary female sex organs** that **produce the female gamete (ovum)** and **several steroid hormones (ovarian hormones)**. They are **located one on each side of the lower abdomen**. **Each ovary is about 2 to 4 cm in length** and is **connected to the pelvic wall and uterus by ligaments**.

Each ovary is **covered by a thin epithelium** which **encloses the ovarian stroma**. The **stroma is divided into two zones -** a **peripheral cortex** and an **inner medulla**. **Ovarian follicles in different stages of development are embedded in the stroma**.

2.2b The Accessory Ducts - Oviducts, Uterus, Cervix, Vagina

The **oviducts (fallopian tubes)**, **uterus** and **vagina** constitute the **female accessory ducts**. **Each fallopian tube is about 10-12 cm long** and **extends from the periphery of each ovary to the uterus**. Its parts, from the ovary towards the uterus, are:

1 **Infundibulum -** the **funnel-shaped part closer to the ovary**. Its edges have **finger-like projections called fimbriae**, which **help in the collection of the ovum after ovulation**.

2 **Ampulla -** the **wider part of the oviduct** that the infundibulum leads into.

3 **Isthmus -** the **last part of the oviduct**, which has a **narrow lumen** and **joins the uterus**.

The **uterus is single** and is also called the **womb**. Its **shape is like an inverted pear**, and it is **supported by ligaments attached to the pelvic wall**. The **uterus opens into the vagina through a narrow cervix**. The **cavity of the cervix is called the cervical canal**, which **along with the vagina forms the birth canal**.

The **wall of the uterus has three layers of tissue**:

Layer	Position	Nature / feature
Perimetrium	External	Thin membranous layer
Myometrium	Middle	Thick layer of smooth muscle; **exhibits strong contraction during delivery of the baby**
Endometrium	Inner	Glandular layer that **lines the uterine cavity** and **undergoes cyclical changes during the menstrual cycle**

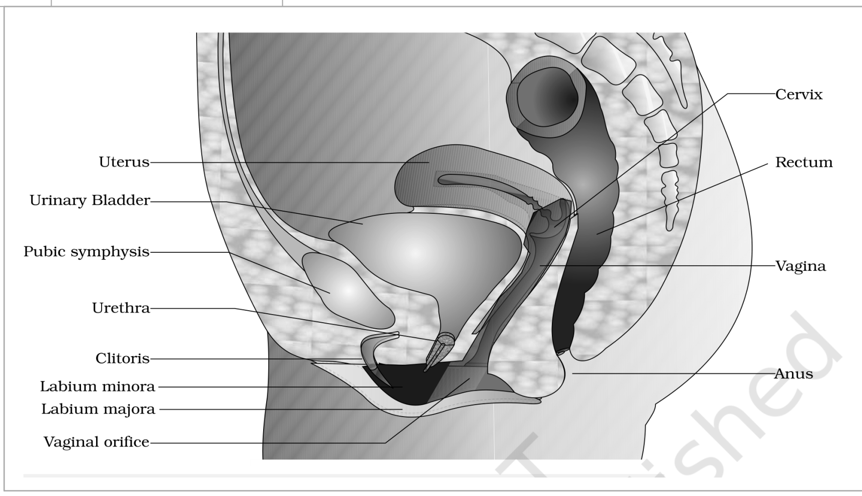

Fig. 2.3a - Diagrammatic sectional view of the female pelvis showing the reproductive system. Labelled parts: uterus, urinary bladder, pubic symphysis, urethra, clitoris, labium minora, labium majora, vaginal orifice, cervix, rectum, vagina and anus.

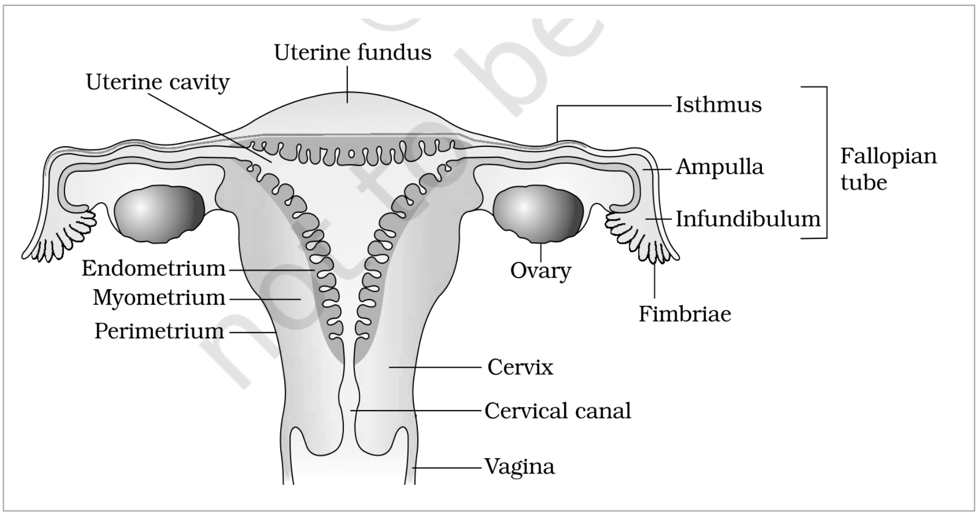

Fig. 2.3b - Diagrammatic sectional view of the female reproductive system. Labelled parts: uterine fundus, uterine cavity, endometrium, myometrium, perimetrium, isthmus, ampulla, infundibulum, fallopian tube, ovary, fimbriae, cervix, cervical canal and vagina.

2.2c The External Genitalia

The **female external genitalia** include the **mons pubis**, **labia majora**, **labia minora**, **hymen** and **clitoris**.

- **Mons pubis -** a **cushion of fatty tissue covered by skin and pubic hair**.

• **Labia majora - fleshy folds of tissue** which **extend down from the mons pubis and surround the vaginal opening**.

• **Labia minora - paired folds of tissue under the labia majora**.

• **Hymen -** a membrane that **often partially covers the opening of the vagina**.

• **Clitoris -** a **tiny finger-like structure** which **lies at the upper junction of the two labia minora above the urethral opening**.

> ⓘ ***[NOTE]*** *The* ***hymen is often torn during the first coitus (intercourse)****. It can* ***also be broken*** *by a* ***sudden fall or jolt****, insertion of a* ***vaginal tampon****, or* ***active participation in some sports like horseback riding, cycling, etc.*** *In some women the* ***hymen persists even after coitus****. Therefore the* ***presence or absence of the hymen is not a reliable indicator of virginity or sexual experience****.*

2.2d The Mammary Glands

A **functional mammary gland is characteristic of all female mammals**, and the **mammary glands are one of the female secondary sexual characteristics**. The **mammary glands are paired structures (breasts)** that **contain glandular tissue and a variable amount of fat**. Milk drains from the gland along a fixed path:

1. The **glandular tissue of each breast is divided into 15-20 mammary lobes** containing **clusters of cells called alveoli**.
2. The **cells of the alveoli secrete milk**, which is **stored in the cavities (lumens) of the alveoli**.
3. The **alveoli open into mammary tubules**.
4. The **tubules of each lobe join to form a mammary duct**.
5. Several **mammary ducts join to form a wider mammary ampulla**, which is **connected to a lactiferous duct** and **opens at the nipple**, from where **milk is expressed during breastfeeding**.

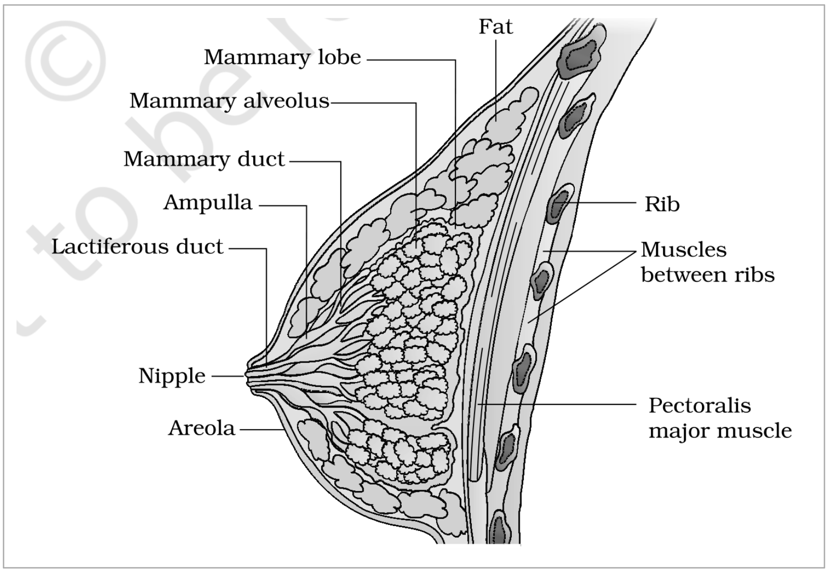

Fig. 2.4 - A diagrammatic sectional view of a mammary gland. Labelled parts: mammary lobe, mammary alveolus, mammary duct, ampulla, lactiferous duct, nipple, areola, fat, rib, muscles between ribs and pectoralis major muscle.

2.3 Gametogenesis

The **primary sex organs -** the **testis in males** and the **ovaries in females - produce gametes**, i.e. **sperms and ovum respectively**, by the process called **gametogenesis**.

2.3a Spermatogenesis

In the testis, **immature male germ cells (spermatogonia) produce sperms by spermatogenesis**, a process that **begins at puberty**. The steps are:

1. **Spermatogonia** (sing. spermatogonium) on the inside wall of the seminiferous tubules **multiply by mitotic division and increase in number**. Each **spermatogonium is diploid and contains 46 chromosomes**.

2 Some spermatogonia, called **primary spermatocytes**, **periodically undergo meiosis**.

3 A **primary spermatocyte completes the first meiotic division (reduction division)**, producing **two equal, haploid secondary spermatocytes**, each with **only 23 chromosomes**.

4 The **secondary spermatocytes undergo the second meiotic division** to produce **four equal, haploid spermatids**.

5 The **spermatids are transformed into spermatozoa (sperms)** by the process called **spermiogenesis**.

6 After spermiogenesis, the **sperm heads become embedded in the Sertoli cells** and are **finally released from the seminiferous tubules** by the process called **spermiation**.

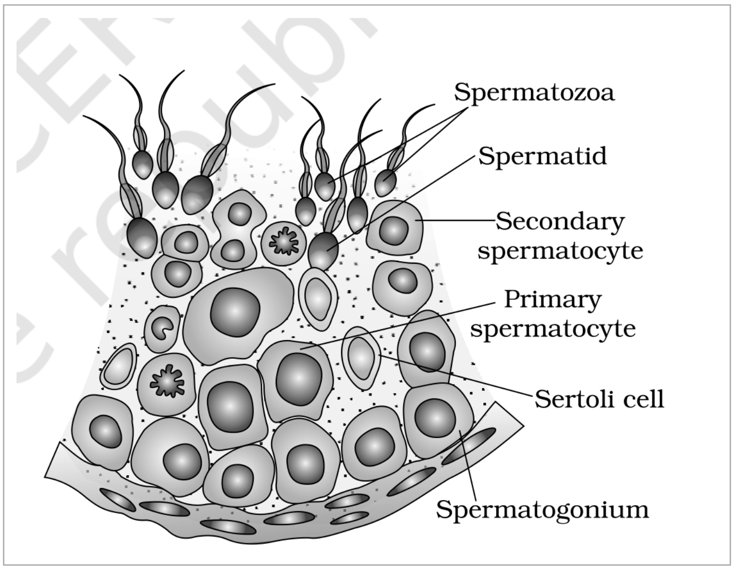

Fig. 2.5 - Diagrammatic sectional view of a seminiferous tubule (enlarged). Labelled cells: spermatogonium, primary spermatocyte, secondary spermatocyte, spermatid, spermatozoa and Sertoli cell.

2.3b Hormonal Control of Spermatogenesis

Spermatogenesis **starts at puberty** due to a **significant increase in the secretion of gonadotropin releasing hormone (GnRH)**, which is a **hypothalamic hormone**.

1 **Increased GnRH acts at the anterior pituitary gland** and **stimulates secretion of two gonadotropins - luteinising hormone (LH)** and **follicle stimulating hormone (FSH)**.

2 **LH acts at the Leydig cells** and **stimulates synthesis and secretion of androgens**. The **androgens, in turn, stimulate the process of spermatogenesis**.

3 **FSH acts on the Sertoli cells** and **stimulates secretion of some factors which help in the process of spermiogenesis**.

2.3c Structure of the Sperm

A **sperm is a microscopic structure** composed of a **head**, a **neck**, a **middle piece** and a **tail**. A **plasma membrane envelops the whole body of the sperm**.

- **Head:** contains an **elongated haploid nucleus**, the **anterior portion of which is covered by a cap-like structure, the acrosome**. The **acrosome is filled with enzymes that help fertilisation of the ovum**.
- **Middle piece:** possesses **numerous mitochondria**, which **produce energy for the movement of the tail** that **facilitates sperm motility essential for fertilisation**.

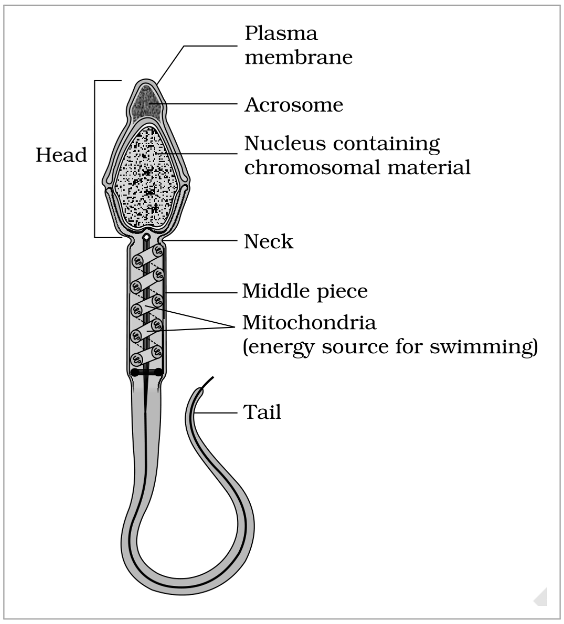

Fig. 2.6 - Structure of a sperm. Labelled parts: plasma membrane, acrosome, nucleus, head, neck, middle piece, mitochondria and tail.

The **human male ejaculates about 200 to 300 million sperms during a coitus**. For **normal fertility**, **at least 60 per cent of sperms must have normal shape and size** and **at least 40 per cent must show vigorous motility**.

Sperms **released from the seminiferous tubules are transported by the accessory ducts**. The **secretions of the epididymis, vas deferens, seminal vesicle and prostate are essential for the maturation and motility of sperms**. The **seminal plasma along with the sperms constitutes the semen**. The **functions of the male sex accessory ducts and glands are maintained by the testicular hormones (androgens)**.

2.3d Oogenesis

Oogenesis - the **formation of a mature female gamete -** is **markedly different from spermatogenesis**. It is **initiated during the embryonic development stage**, when a **couple of million gamete mother cells (oogonia) are formed within each fetal ovary**. **No more oogonia are formed and added after birth**.

1. **Oogonia start division and enter prophase-I of the meiotic division**, then get **temporarily arrested at that stage**; they are now called **primary oocytes**.
2. Each **primary oocyte gets surrounded by a layer of granulosa cells** and is called a **primary follicle**. A **large number of these follicles degenerate during the phase from birth to puberty**, so that **at puberty only 60,000-80,000 primary follicles are left in each ovary**.
3. The **primary follicles get surrounded by more layers of granulosa cells and a new theca** and are called **secondary follicles**.
4. A secondary follicle **transforms into a tertiary follicle**, characterised by a **fluid-filled cavity called the antrum**. The **theca layer is organised into an inner theca interna and an outer theca externa**.
5. At the **tertiary-follicle stage the primary oocyte grows in size and completes its first meiotic division**. This is an **unequal division**, resulting in a **large haploid secondary oocyte and a tiny first polar body**; the **secondary oocyte retains the bulk of the nutrient-rich cytoplasm of the primary oocyte**.
6. The **tertiary follicle further changes into the mature follicle or Graafian follicle**. The **secondary oocyte forms a new membrane called the zona pellucida** surrounding it.
7. The **Graafian follicle ruptures to release the secondary oocyte (ovum) from the ovary** by the process called **ovulation**.

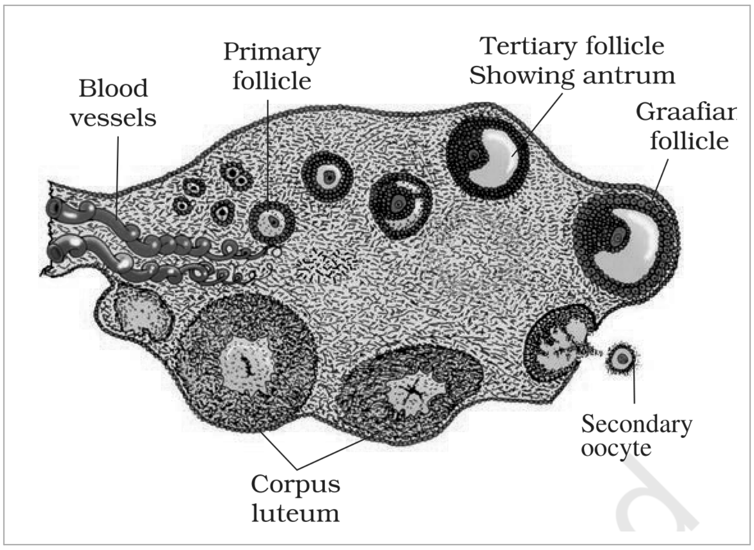

Fig. 2.7 - Diagrammatic sectional view of an ovary. Labelled parts: blood vessels, primary follicle, tertiary follicle, antrum, Graafian follicle, secondary oocyte and corpus luteum.

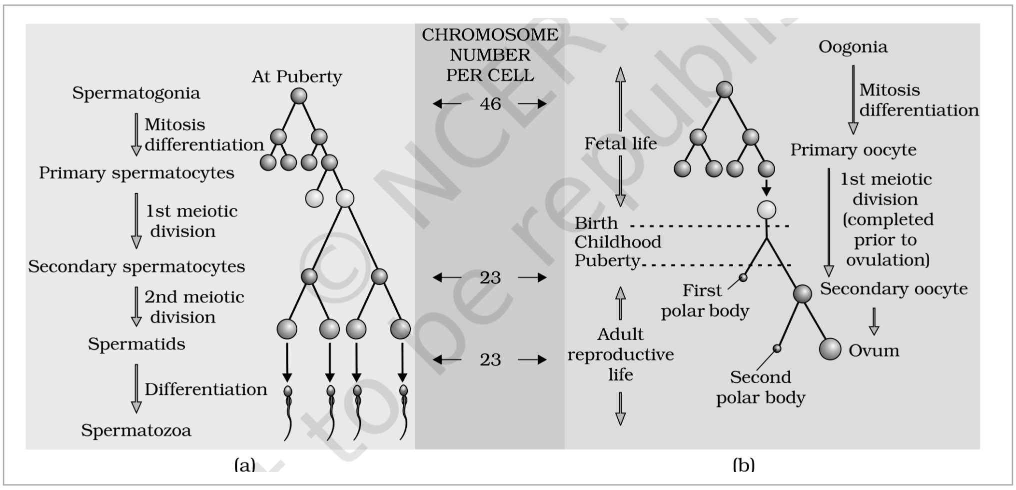

Fig. 2.8 - Schematic representation of (a) spermatogenesis and (b) oogenesis. Labelled: spermatogonia, mitosis, primary spermatocytes, secondary spermatocytes, spermatids and spermatozoa, with the chromosome number per cell shown as 46 and 23; and oogonia, primary oocyte, secondary oocyte, first polar body, second polar body and ovum.

2.4 Menstrual Cycle

The **reproductive cycle in the female primates** (e.g. **monkeys, apes and human beings**) is called the **menstrual cycle**. The **first menstruation begins at puberty** and is called **menarche**.

In human females, **menstruation is repeated at an average interval of about 28/29 days**. The **cycle of events starting from one menstruation till the next is called the menstrual cycle**, and **one ovum is released (ovulation) during the middle of each menstrual cycle**.

The cycle proceeds through the following phases:

1. **Menstrual phase:** the cycle **starts with the menstrual phase**, when **menstrual flow occurs**; it **lasts for 3-5 days**. The **menstrual flow results due to breakdown of the endometrial lining of the uterus and its blood vessels**, which forms a **liquid that comes out through the vagina**.
2. **Follicular phase:** the **primary follicles grow to become a fully mature Graafian follicle** and, **simultaneously, the endometrium regenerates through proliferation**.
3. **Ovulatory phase:** the **LH surge induces rupture of the Graafian follicle and release of the ovum (ovulation)**.
4. **Luteal phase:** the **remaining parts of the Graafian follicle transform into the corpus luteum**, which **secretes large amounts of progesterone**.

[NOTE] Menstruation only occurs if the released ovum is not fertilised. *A* ***lack of menstruation may be indicative of pregnancy****; it* ***may also be caused due to some other underlying causes like stress, poor health, etc.***

These **changes in the ovary and the uterus are induced by changes in the levels of pituitary and ovarian hormones**. The **secretion of gonadotropins (LH and FSH) increases gradually during the follicular phase** and **stimulates follicular development as well as the secretion of estrogens by the growing follicles**. **Both LH and FSH attain a peak level in the middle of the cycle (about the 14th day)**. The **rapid secretion of LH leading to its maximum level during mid-cycle, called the LH surge, induces rupture of the Graafian follicle and release of the ovum (ovulation)**.

The **progesterone secreted by the corpus luteum is essential for maintenance of the endometrium**. Such an **endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy**. **During pregnancy all events of the menstrual cycle stop and there is no menstruation**. **In the absence of fertilisation, the corpus luteum degenerates**, **causing disintegration of the endometrium leading to menstruation**, which marks a **new cycle**.

In human beings, **menstrual cycles cease around 50 years of age**; that is termed **menopause**. **Cyclic menstruation is an indicator of a normal reproductive phase** and **extends between menarche and menopause**.

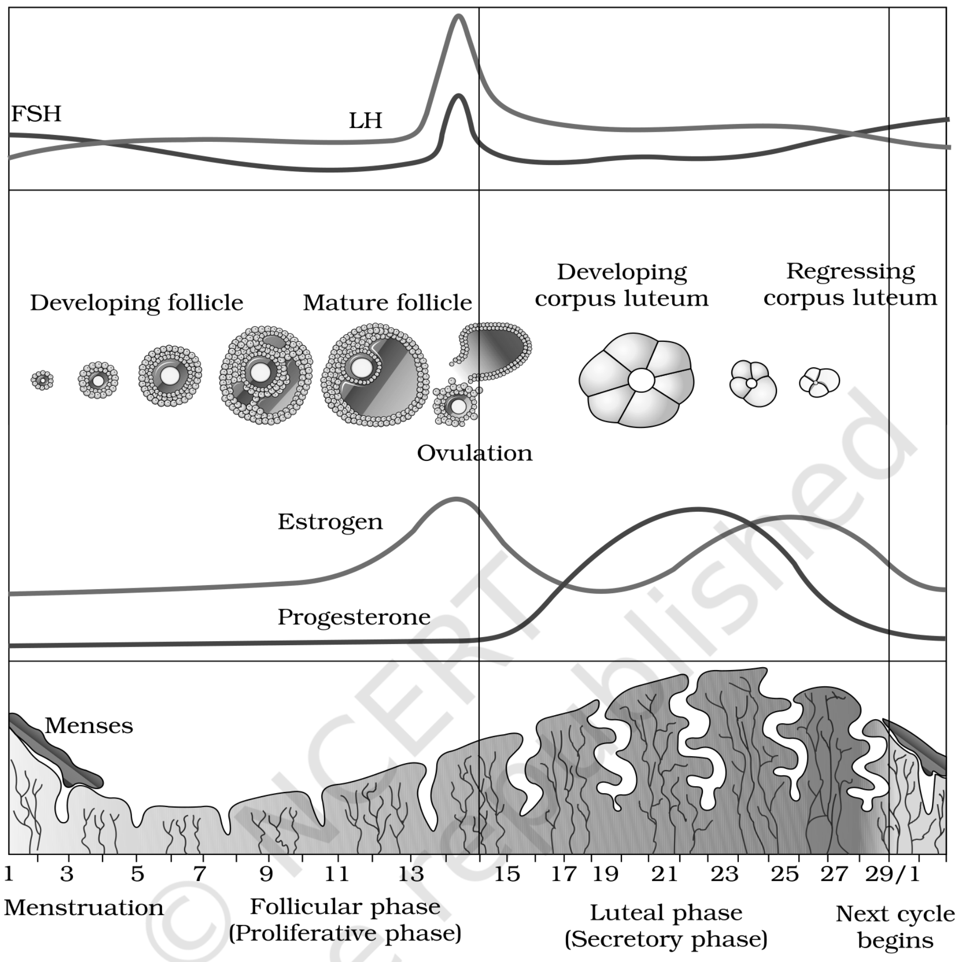

Fig. 2.9 - Diagrammatic presentation of the various events during a menstrual cycle. Labelled: FSH, LH, developing follicle, mature follicle, corpus luteum, ovulation, estrogen, progesterone, menses, the follicular phase and the luteal phase.

[NOTE] Menstrual Hygiene. *Maintenance of hygiene and sanitation during menstruation is very important. Take a bath and clean regularly; use* ***sanitary napkins*** *or* ***clean homemade pads****; change them* ***after every 4-5 hrs*** *as per requirement; dispose of used napkins by wrapping them in used paper;* ***do not throw used napkins in the drainpipe of toilets or in the open area****; and after handling the napkin,* ***wash hands with soap****.*

2.5 Fertilisation and Implantation

During **copulation (coitus)**, **semen is released by the penis into the vagina (insemination)**. The **motile sperms swim rapidly, pass through the cervix, enter the uterus and finally reach the ampullary region of the fallopian tube**. The **ovum released by the ovary is also transported to the ampullary region, where fertilisation takes place**.

> ⓘ ***[NOTE] Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.*** *This is* ***why not all copulations lead to fertilisation and pregnancy****.*

2.5a Fertilisation

The **process of fusion of a sperm with an ovum is called fertilisation**. During fertilisation, a **sperm comes in contact with the zona pellucida layer of the ovum** and **induces changes in the membrane that block the entry of additional sperms**. This **ensures that only one sperm can fertilise an ovum**. The **secretions of the acrosome help the sperm enter the cytoplasm of the ovum through the zona pellucida and the plasma membrane**.

This entry **induces completion of the meiotic division of the secondary oocyte**. The **second meiotic division is also unequal** and **results in a second polar body and a haploid ovum (ootid)**. The **haploid nucleus of the sperm and that of the ovum fuse together to form a diploid zygote**.

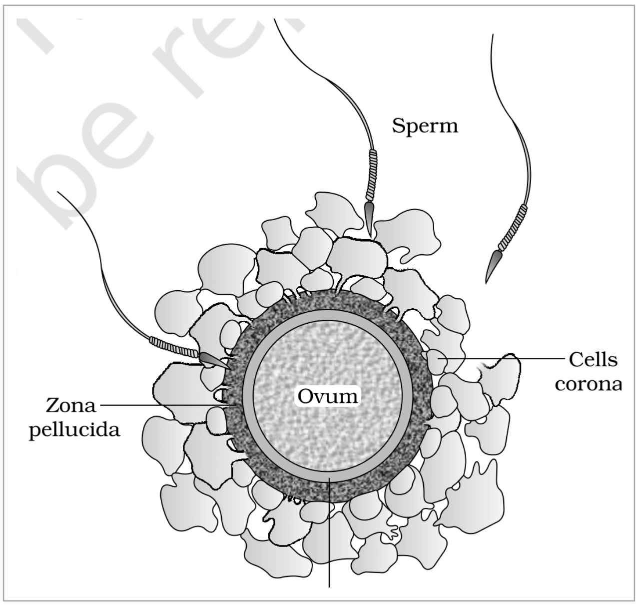

Fig. 2.10 - Ovum surrounded by a few sperms. Labelled parts: sperm, zona pellucida, corona radiata, perivitelline space and ovum.

2.5b Sex Determination

The **sex of the baby is decided at the fertilisation stage**. The **chromosome pattern in the human female is XX and in the male is XY**. **All haploid gametes (ova) produced by the female have the sex chromosome X**, whereas in **male gametes (sperms) the sex chromosome could be either X or Y - 50 per cent of sperms carry X and the other 50 per cent carry Y**.

Thus the **zygote carries either XX or XY depending on whether the X- or Y-bearing sperm fertilised the ovum**: **XX develops into a female baby and XY into a male** (more on chromosomal patterns in Chapter 5). Therefore, **scientifically it is correct to say that the sex of the baby is determined by the father and not by the mother**.

2.5c Cleavage and Implantation

1. **Mitotic division called cleavage starts as the zygote moves through the isthmus of the oviduct towards the uterus** and **forms 2, 4, 8, 16 daughter cells called blastomeres**.
2. The **embryo with 8 to 16 blastomeres is called a morula**.

3 The **morula continues to divide and transforms into a blastocyst as it moves further into the uterus**. The **blastomeres in the blastocyst are arranged into an outer layer called the trophoblast and an inner group of cells attached to the trophoblast called the inner cell mass**.

4 The **trophoblast layer gets attached to the endometrium and the inner cell mass gets differentiated as the embryo**.

5 After attachment, the **uterine cells divide rapidly and cover the blastocyst**, so the **blastocyst becomes embedded in the endometrium** - this is called **implantation**, and it **leads to pregnancy**.

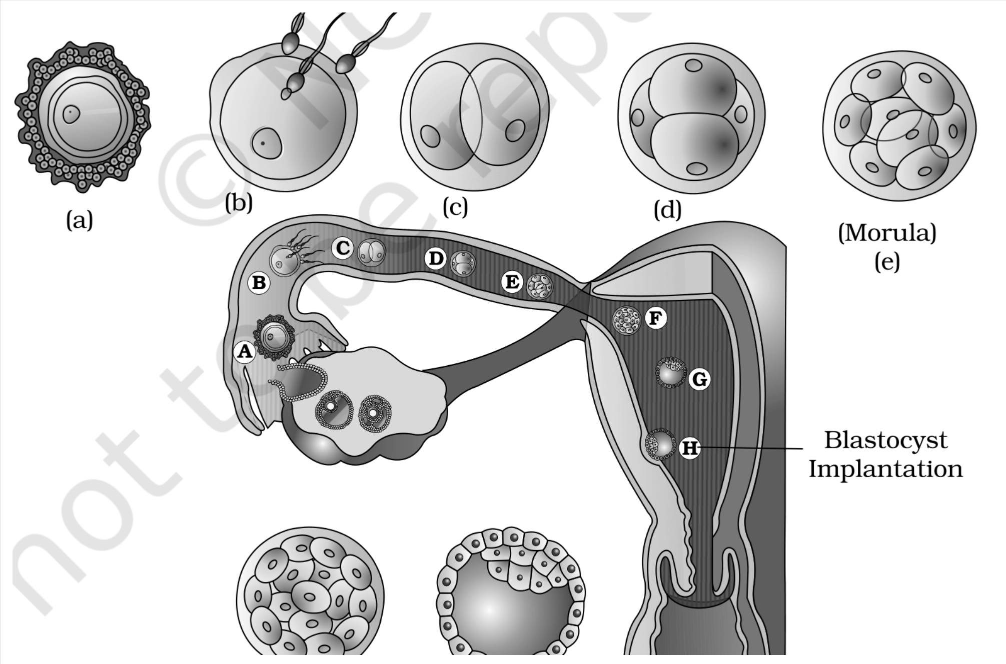

Fig. 2.11 - Transport of the ovum, fertilisation, and passage of the growing embryo through the fallopian tube. Labelled stages: morula, blastocyst and implantation.

2.6 Pregnancy and Embryonic Development

After implantation, **finger-like projections appear on the trophoblast called chorionic villi**, which are **surrounded by the uterine tissue and maternal blood**. The **chorionic villi and uterine tissue become interdigitated and jointly form a structural and functional unit between the developing embryo (foetus) and the maternal body called the placenta**.

The **placenta facilitates the supply of oxygen and nutrients to the embryo** and the **removal of carbon dioxide and excretory/waste materials produced by the embryo**. It is **connected to the embryo through an umbilical cord**, which **helps in the transport of substances to and from the embryo**.

The **placenta also acts as an endocrine tissue** and **produces hormones like human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens, progestogens, etc.** In the **later phase of pregnancy, a hormone called relaxin is also secreted by the ovary**.

> *[NOTE]* ***hCG, hPL and relaxin are produced in women only during pregnancy.*** *In addition,* ***during pregnancy the levels of other hormones like estrogens, progestogens, cortisol, prolactin, thyroxine, etc. are increased several-fold in the maternal blood****. The* ***increased production of these hormones is essential for supporting fetal growth, metabolic changes in the mother, and maintenance of pregnancy****.*

Immediately after implantation, the inner cell mass (embryo) differentiates into an outer layer, the ectoderm, and an inner layer, the endoderm; a **mesoderm soon appears between them**. These **three (germ) layers give rise to all tissues (organs) in adults**. The **inner cell mass contains certain cells called stem cells** which have the **potency to give rise to all the tissues and organs**.

Human pregnancy lasts 9 months. The growth of the foetus follows a recognisable timeline:

Stage of pregnancy	Development observed
After **1 month**	The **embryo's heart is formed**; the **first sign of the growing foetus may be noticed by listening to the heart sound through a stethoscope**.
End of **2 months**	The **foetus develops limbs and digits**.
End of **12 weeks (first trimester)**	**Most major organ systems are formed**; the **limbs and external genital organs are well-developed**.
Fifth month	The **first movements of the foetus and the appearance of hair on the head are usually observed**.
End of about **24 weeks (end of second trimester)**	The **body is covered with fine hair, the eye-lids separate, and eyelashes are formed**.
End of **9 months**	The **foetus is fully developed and ready for delivery**.

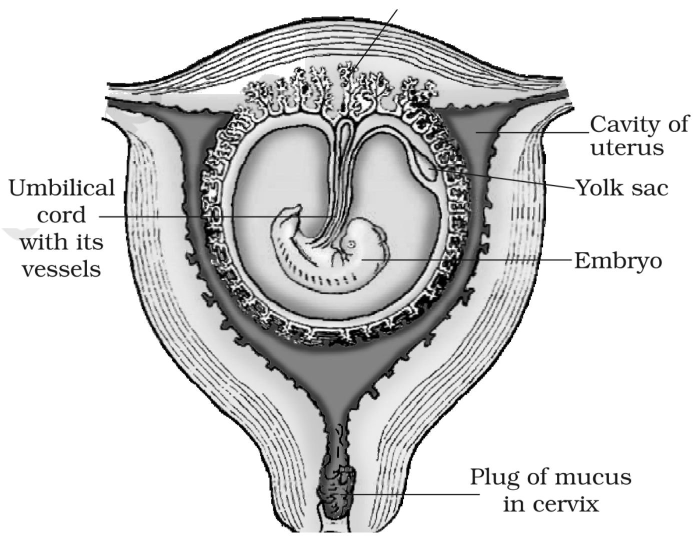

Fig. 2.12 - The human foetus within the uterus. Labelled parts: placental villi, umbilical cord, cavity of uterus, yolk sac, embryo and cervix.

2.7 Parturition and Lactation

The **average duration of human pregnancy is about 9 months**, which is called the **gestation period**. **Vigorous contraction of the uterus at the end of pregnancy causes expulsion/delivery of the foetus**. The **process of delivery of the foetus (childbirth) is called parturition**.

Parturition is induced by a complex neuroendocrine mechanism involving **cortisol, estrogens and oxytocin**. It proceeds as a self-reinforcing reflex:

1. **Signals for parturition originate from the fully developed foetus and the placenta** and **induce mild uterine contractions called the foetal ejection reflex**.
2. The **foetal ejection reflex triggers the release of oxytocin from the maternal pituitary**.
3. **Oxytocin acts on the uterine muscle and causes stronger uterine contractions**, which in turn **stimulate further secretion of oxytocin**.
4. This **stimulatory reflex between uterine contraction and oxytocin secretion continues**, resulting in **stronger contractions and expulsion of the baby through the birth canal (parturition)**.

Soon after the infant is delivered, the placenta is also expelled out of the uterus.

The **mammary glands undergo differentiation during pregnancy** and **start producing milk towards the end of pregnancy by the process called lactation**. **Lactation helps the mother in feeding the new-born.** The **milk produced during the initial few days of lactation is called colostrum**, which **contains several antibodies absolutely essential to develop resistance for the new-born babies**. Hence **breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby**.

Quick Recap

- Humans are **sexually reproducing and viviparous**. The male reproductive system is a **pair of testes**, the **male sex accessory ducts, accessory glands and external genitalia**.
- Each testis has **about 250 testicular lobules**, each with **one to three seminiferous tubules** lined by **spermatogonia (which undergo meiosis to form sperms) and Sertoli cells (which nourish the germ cells)**; the **Leydig cells outside the tubules secrete androgens**. The male external genitalia is the **penis**.
- The female reproductive system is a **pair of ovaries, a pair of oviducts, a uterus, a vagina, external genitalia and a pair of mammary glands**. The **ovaries produce the ovum and ovarian (steroid) hormones**, with **ovarian follicles at different stages embedded in the stroma**. The **uterus has three layers - perimetrium, myometrium and endometrium**, and the **mammary glands are a female secondary sexual characteristic**.
- **Spermatogenesis** yields sperms carried by the accessory ducts; a sperm has a **head, neck, middle piece and tail**. **Oogenesis** forms the mature female gamete. The **reproductive cycle of female primates is the menstrual cycle**, which starts only after puberty, releases **one ovum per cycle**, and is **controlled by pituitary and ovarian hormones**.
- After coitus, **sperms reach the ampulla, where a sperm fertilises the ovum to form a diploid zygote**; the **X- or Y-bearing sperm determines the sex**. The zygote forms a **blastocyst that is implanted in the uterus**, the **placenta connects foetus and mother**, and **gestation lasts about 9 months**.
- **Parturition is induced by a complex neuroendocrine mechanism involving cortisol, estrogens and oxytocin**; the **mammary glands secrete milk (lactation) after childbirth**, beginning with the antibody-rich **colostrum**.

Terms Used in the Exercises

Two exercise questions rely on ideas the chapter itself does not spell out. Both are answered here using only chapter-based reasoning.

- **Identical vs fraternal twins.** The chapter states that **one ovum is normally released per menstrual cycle** and that a **single sperm fertilises a single ovum** to form one zygote. **Identical twins** arise when **one fertilised ovum (one zygote) splits** to give two embryos - they share the same genetic make-up. **Fraternal (non-identical) twins** arise when **two separate ova are released and fertilised by two separate sperms**, giving two genetically different embryos.
- **Eggs released for a litter of six puppies.** Because each puppy develops from its own fertilised ovum (one zygote per offspring), a dog delivering a **litter of six puppies must have released at least six ova** in that cycle, each fertilised by a separate sperm.

Every fact, number, name, qualifier, table row, figure and figure label in NCERT Class 12 Chapter 2 is carried above. Nothing outside the source chapter has been added, except the clearly marked NOTE/MEMORY AID material and the exercise-gap explanations, which are derived only from chapter content.